Market Research and Comprehensive Gap Analysis in Computational Electromagnetics Educational Material

The discipline of computational electromagnetics (CEM) has reached a pivotal juncture in its seventy-five-year history, a period fundamentally defined by the co-evolution of the IEEE Antennas and Propagation Society (AP-S) and the exponential growth of computing hardware capabilities.[1] As the complexity of electronic systems has advanced from simple radar components to the intricate, multi-scale architectures required for the Internet of Things (IoT) and 5G/6G communications, the demand for sophisticated simulation tools has continuously pushed the boundaries of available computer resources.[3] However, market research and academic surveys indicate a growing chasm between the capabilities of modern commercial field solvers and the pedagogical frameworks used to train the engineers who operate them.[5] This report identifies critical gaps in CEM educational materials, focusing on under-represented topics, poorly explained methodologies, and the misalignment between academic instruction and industrial workforce requirements.

Historical Evolution and the Paradox of Progress

The history of CEM can be categorized into distinct eras, each driven by a specific technological paradigm. In the early mainframe era, spanning from the 1950s to the 1970s, numerical analysis was largely secondary to analytical derivations, and the founding fathers of modern electromagnetic theory—Maxwell, Hertz, and Heaviside—provided the theoretical bedrock for what was then a nascent computational field.[1] The founding of the AP-S in 1949 as the IRE Professional Group on Antennas and Propagation marked the beginning of a formalized effort to bridge physics and engineering through systematic computation.[2]

As high-level programming languages such as Fortran, ALGOL, and later C became widely available, the transition from analytical techniques to systematic numerical solutions like the Finite Element Method (FEM) and the Method of Moments (MoM) began in earnest.[2] By the 1990s, the "state of the art" in simulations had moved from electrically small problems to sophistication in geometry and material modeling, largely driven by advances in computational horsepower.[4] Today, computational electromagnetics is a routine part of the design ecosystem, yet this success has introduced a paradox: the more "user-friendly" the software becomes, the less the user understands the physics governing the simulation.[4]

The Evolution of Computational Paradigms in Electromagnetics

Era	Dominant Hardware	Core Educational Focus	Primary Numerical Milestone
Foundations (1949-1965)	Mainframes/Vacuum Tubes	Analytical derivations, vector calculus	Initial use of computers for integral equations [1]
Early Numerical (1966-1985)	Transistorized systems	Introduction of FEM and FDTD theory	Commercialization of basic field solvers [2]
Modern Simulation (1986-2010)	Workstations/Beowulf Clusters	Software GUI training, CAD integration	Large-scale parallelization and MLFMM [4]
Cognitive/AI Era (2011-Present)	GPUs/Cloud/Quantum Accelerators	AI/ML surrogates, multi-physics	Integration of Deep Learning and QEM [8]

The transition into the current era has been marked by a shift in how engineers validate their work. Historically, measurements were the gold standard for verifying theory; now, designers frequently ask whether their measured data agrees with a commercial solver like HFSS.[4] This reversal underscores the absolute necessity for educational materials that emphasize critical evaluation over rote operation.

The Black-Box Syndrome and the Erosion of Physical Intuition

A primary gap identified in current CEM education is the increasing reliance on "black-box" software. Market analysis of educational programs reveals a tendency for students to treat commercial field solvers as infallible tools rather than as discrete approximations of physical reality.[5] This reliance is often encouraged by the "button-clicking" nature of modern GUIs, which allow users to set up complex simulations without understanding the underlying discretization or the limitations of the chosen numerical method.[9]

The psychological impact of this trend is significant. Students often lack the critical thinking required to interpret results when a simulation fails to converge or produces non-physical data, such as field spikes or violation of energy conservation.[7] In professional forums, this is often described as the "lie to yourself" phenomenon, where users trust software defaults—such as a specific element type in FEM—without realizing that those defaults may be obsolete or inappropriate for the geometry at hand.[9] For example, a solver defaulting to first-order tetrahedral elements can lead to outrageously inaccurate results in high-frequency

applications, yet educational materials often fail to highlight these nuances.[9]

Pedagogical Critiques of Current CEM Instruction

Identified Gap	Underlying Mechanism	Impact on Engineering Quality
Reliance on Black-Boxes	Excessive focus on GUI navigation over algorithm theory	Inability to troubleshoot or verify simulation failure [7]
Disconnect from Physics	Treating CEM as a software skill rather than applied physics	Erosion of intuitive understanding of wave behavior [10]
Lack of V&V Skills	Omission of formal Validation and Verification procedures	Overconfidence in simulation fidelity without proof [7]
Default Trust	Failure to critique software default settings (e.g., meshing)	Systematic error accumulation in complex designs [9]

Evidence from student-centered electromagnetic instruction suggests that the abstract nature of electromagnetics, coupled with traditional, derivation-heavy methods, has led to a decline in interest in RF and microwave engineering.[10] There is a clear need for educational interventions that decouple physical understanding from pure mathematics through the use of high-quality animations, visualizations, and simulation-aided learning.[10] Resources such as EMPossible have gained widespread acclaim precisely because they fill this gap, providing clear, visual explanations of how math equations transform into computer codes.[13]

Under-Represented Numerical Methods and Advanced Techniques

While Finite-Difference Time-Domain (FDTD) and the Finite Element Method (FEM) are the staples of most CEM curricula, several advanced techniques are poorly explained or entirely omitted from standard educational materials. This creates a workforce that is ill-equipped to select the most efficient tool for specific high-performance applications.

The Discontinuous Galerkin Time-Domain (DGTD) Method

DGTD is identified as a significantly under-represented method in mainstream CEM education. It offers a hybrid approach that combines the spatial explicit algorithms of FDTD with the geometric flexibility of FEM.[14] DGTD is particularly effective for transient multiscale electromagnetic problems because it allows for high-order (p-) accuracy and superior meshing (h-) refinement, which are intrinsic disadvantages of traditional FDTD solvers.[15]

Despite its advantages, DGTD is rarely taught in depth. In simulations of high-Q optical resonators, DGTD outperforms FDTD by avoiding the phase errors and "staircasing" approximations that limit the geometric size analyzable with given hardware.[15] The educational gap in DGTD prevents students from understanding how to leverage triangular prismatic space discretization, which is optimal for planar geometries like optical metasurfaces.[16]

The Finite Integration Technique (FIT)

The Finite Integration Technique is another methodology that suffers from a lack of dedicated educational material. Often presented merely as an extension of FDTD, FIT actually provides a more coherent approach for the discrete representation of Maxwell's equations on spatial grids.[17] FIT employs the integral form of Maxwell's equations rather than the differential form, producing Maxwell's Grid Equations (MGEs) that guarantee physical correctness and unique solutions.[17]

Market research into commercial tools shows that FIT is the core technology for prominent solvers like CST Studio Suite, particularly for broadband transient analysis.[18] However, the educational resources for FIT are sparse compared to FEM. This leaves users unable to understand the "speed advantage" FIT provides when solving fast transient and complex structures using integral quantities as state variables.[19]

Differential Forms and Textured Space

At the most theoretical level, the use of Differential Forms (DF) in CEM is an almost entirely absent topic in standard engineering textbooks. Differential forms offer a "textured" view of space, suggesting that space has structure that is often ignored in classical calculus.[20] According to this theory, electromagnetic fields like B and H live in different parts of a textured space and cannot be connected point-wise as in continuous calculus.[20]

Teaching CEM through differential forms allows for a more natural discretization of the finite integration type and facilitates relativistic formulations in four dimensions.[21] However, because this requires a shift from traditional vector analysis to a more geometric perspective, it is often viewed as too abstract for engineering students, despite its potential to provide deeper insights into the nature of electromagnetic fields.[20]

The Computational Literacy Deficit: HPC and

Parallelization

As electromagnetic environments grow in complexity—with features ranging from nanometers on-chip to millimeters on-package—the computational burden has increased exponentially.[3] This necessitates a level of "computational literacy" that is currently missing from many engineering programs.[7]

Computational literacy in the context of CEM involves more than just programming; it requires an understanding of algorithmic complexity, memory management, and the exploitation of High-Performance Computing (HPC) and GPU acceleration.[7] Many practical applications remain out of range for present-day algorithms unless they are optimized for massively parallel digital computers capable of tera-, peta-, or even exa-flop throughputs.[22]

Parallel Programming Gaps in CEM

Skill Area	Practical Requirement	Current Educational Status
Many-Core Optimization	Mapping computations to parallel hardware (GPUs/CPUs)	Rarely integrated into CEM core courses [7]
Algorithmic Patterns	Utilizing CUDA or OpenMP for matrix filling and factoring	Relegated to specialized computer science electives [21]
Matrix Compression	Implementing ACA or H-matrix algorithms for large systems	Advanced topic with limited accessible training [23]
Performance Profiling	Identifying bottlenecks in memory bandwidth or cache	Often absent from engineering methodology [24]

Institutions like TU Darmstadt have recognized this gap, introducing modules that transition students from scripting languages like MATLAB to parallel programming in C++ using the "Eigen" library and OpenMP.[21] However, this is not yet the global standard. The result is a cohort of engineers who can design an antenna but cannot optimize the simulation code to handle a large-scale phased array analysis, which might otherwise take days to complete on a single beam angle.[18]

Mesh Generation and the Science of Discretization

The process of mesh generation is the "heart of finite element analysis" and is a critical step that significantly influences accuracy and efficiency.[26] Yet, market research indicates that mesh generation is one of the most poorly explained topics in CEM educational material.[27]

Meshing is where "science meets intuition".[26] It requires an understanding of the physics involved—such as the skin effect in conductors or field singularities at sharp edges—to determine appropriate meshing resolutions.[28] Educational materials often fail to teach the "geometric and topological considerations" of 3D meshing, such as why a solver might produce "uselessly cramped" nodes in narrowing cross-sections or how badly shaped elements cause slow convergence.[28]

Under-Represented Meshing Techniques

1. **Non-Matching Meshes:** Strategies for discretizing partial differential equations on independent, overlapping meshes using formulations like Nitsche's method are almost entirely absent from introductory texts.[29]
2. **Adaptive Mesh Refinement:** While many commercial tools offer "automatic" refinement, the educational material on how to perform manual p-, h-, or hp-refining adaptively to improve a specific "Quantity of Interest" (QoI) is lacking.[30]
3. **High-Quality Metrics:** Techniques like Delaunay triangulation and the advancing front method are often mentioned but rarely explained in terms of their "quality metrics"—aspect ratio, smoothness, and alignment—which ensure optimal element shape for accurate results.[31]

The failure to teach these "hand-on" skills results in practitioners who rely on automated tools that might sacrifice accuracy for speed, or worse, generate meshes that are "fucking useless in actual application" because they fail to capture the underlying physics.[9]

The Multidisciplinary Barrier: Multi-physics and System-Level Performance

Current CEM education often treats electromagnetics as an isolated subject, yet the industry is increasingly demanding a multidisciplinary approach.[7] Modern high-frequency design requires the simultaneous modeling of coupled electrical, mechanical, and thermal responses.[3]

The signal and power integrity (SPI) of hardware systems—which must be wireless, low-power, and ultra-thin—is essential for IoT and Big Data technologies.[3] This requires CEM practitioners to bridge the gap between abstract equations and the discretization processes used in multi-physics models.[7] However, there is a noted gap in teaching how CEM interacts with

thermal analysis, structural mechanics, and circuit co-simulation.[7]

For instance, the behavior of materials at high frequencies and the packaging of new circuit architectures present "grand challenges" that are beyond the reach of standard simulation capabilities taught in universities.[4] Engineers must often trade accuracy for computational cost, but without a multidisciplinary background, they lack the "informed engineering judgment" needed to make these tradeoffs effectively.[3]

Multi-physics Integration Trends

Integrated Domain	Educational Relevance	Industrial Application
Thermal Coupling	Managing heat dissipation in high-power RF devices	Packaging for wearable and implantable sensors [3]
Mechanical Integrity	Structural responses of thin-film antennas	Driverless vehicles and threat detection scanners [4]
Circuit Co-simulation	Coupled field-circuit modeling for radiated interference	Signal and Power Integrity (SPI) in IoT hardware [3]
Quantum Effects	Quantization of fields in dispersive media	Superconducting qubits and quantum sensors [33]

AI and Machine Learning in Computational Electromagnetics

The confluence of improved hardware performance and advanced network algorithms has driven a surge in the application of artificial intelligence and machine learning (AI/ML) to CEM.[3] This is identified as a "universal priority gap" by both industry executives and engineering talent.[34]

AI and ML are increasingly common in numerical modeling for their capability to self-extract numerical models from behavioral examples and provide "prompt evaluations" of electromagnetic fields.[8] This is particularly useful in optimized design and the iterative resolution of inverse problems, where traditional numerical models may be too computationally expensive.[8]

Under-Represented Topics in AI/ML for CEM

1. **Physics-Informed Neural Networks (PINNs):** While conventional AI models are often used as "black-boxes," there is a lack of educational material on how to constrain these models using partial differential equations (PDEs).[8]
2. **Surrogate Modeling:** Training AI to act as a fast surrogate for full-wave solvers is a high-demand skill, yet the methodologies for finding a "good match between input data and network architecture" are poorly documented in a CEM context.[8]
3. **Differentiable Ray Tracing:** This emerging technique offers a real alternative to conventional deep learning for radio propagation modeling, but it is not yet a part of the standard curriculum.[35]

Surveys indicate that 70% of executives in industrial manufacturing and 66% in technology engineering see AI/ML as a critical skills gap.[34] However, workers are often skeptical of being replaced by AI, even while executives emphasize that the failure to embrace AI poses a greater risk to competitiveness.[34]

Quantum Electromagnetics: The Future Gap

The quantization of the electromagnetic field is a subject that is currently gaining momentum but is almost entirely under-represented in classical engineering education.[32] Quantum Electromagnetics (QEM) involves the application of quantum theory to provide a better understanding of effects in technologies like quantum computers, sensors, and communication systems.[32]

Classically-trained electromagnetic engineers have a vital role to play in making these quantum technologies a reality, yet the foundational concepts—such as the vector-scalar potential (A-Φ) formulation in the generalized Lorenz gauge—are rarely taught from a quantum perspective.[32] This results in a "small and aging workforce" that is ill-prepared for competitive research directions in spin-based quantum computing and device characterization.[36]

Quantum Electromagnetics Educational Needs

- **Quantization in Inhomogeneous Media:** Techniques for quantizing fields in reciprocal, anisotropic, and inhomogeneous media are virtually absent from standard literature.[32]
- **Coupled Field-Transmon Modeling:** The interaction of electromagnetic fields with superconducting circuit qubits requires new mathematical descriptions suited for numerical solvers rather than simple circuit element descriptions.[32]
- **Numerical Algorithms for QEM:** There is a need for numerical algorithms that can model quantum electromagnetic effects in dispersive media, which is a prerequisite for advanced quantum sensors.[33]

The "Harrington-Mittra Award in Computational Electromagnetics" and the "Chen-To Tai

Distinguished Educator Award" from the AP-S are beginning to recognize achievement in these areas, but the transition into mainstream curriculum development remains slow.[38]

The Industry-Academia Skills Gap and the "Valley of Death"

The disconnect between academic research and industrial application is often described as the "Valley of Death".[6] Despite the wealth of cutting-edge research produced in universities, it is difficult for industry to extract these learnings and embed them into day-to-day operational activities.[6]

Market research identifies several factors contributing to this gap:

- **Misaligned Objectives:** Academia focuses on citations and theoretical outputs, whereas industry focuses on financial KPIs and engagement.[6]
- **Timeline Mismatch:** Academic projects often span years, while industry requires a turnover in months.[6]
- **Reproducibility Crisis:** Methodologies in academic papers are often not clear enough to enable reproduction in an industrial setting, lacking open data and code.[6]
- **IT and Coding Gaps:** There is a "desperate need" for graduates who are proficient in version control, code review, and applying the latest programming approaches—skills that are often neglected in favor of modeling and advanced algorithms.[6]

Survey of Industrial Skills Gaps

Skill Category	Executive Perception of Gap	Worker Perception of Gap	Impact on Hiring
AI/ML/Data Science	70% (High)	56% (Moderate)	Primary barrier to high-speed design roles [34]
Process Automation	40% (Moderate)	33% (Low)	Slows industrial manufacturing innovation [34]
Cybersecurity (RF)	46% (Moderate)	35% (Moderate)	Essential for smart infrastructure defense [34]

Soft Skills	32% (Moderate)	20% (Low)	Impacts leadership and team collaboration [39]

Industry reports suggest that one in three engineering roles in the U.S. goes unfilled, with electrical engineering being a main contributor to this gap.[40] Enrollment in electrical engineering degrees has declined significantly compared to computer science, partly due to a "negative image" and a perceived lack of excitement among younger generations.[40]

Market Trends in CEM Educational Resources

The K-12 and professional instructional material market is expected to grow to $37.1 billion by 2030, driven by a shift toward digital-first models and AI-enabled content.[41] In the specialized field of CEM, this is manifesting as a move away from traditional textbooks and toward interactive, high-quality curriculum.[43]

Emerging Educational Material Formats

1. **Open Access Journals:** Platforms like the IEEE Journal of Selected Topics in Electromagnetics, Antennas and Propagation (JSTEAP) are fostering "open science" by expanding the availability of data and code sharing.[45]
2. **MOOCs and YouTube:** Channels like "EMPossible" and university platforms like NPTEL (IIT Madras) are filling the gap for visual learners who find traditional textbooks "boring".[13]
3. **Hands-on Training Kits:** The development of mmWave RF front-end beamforming kits and "Wireless Education Kits" provides invaluable resources for rapid prototyping and algorithm testing.[33]
4. **Special Issues:** Professional magazines are increasingly focusing on themes like "Computational Reproducibility and Benchmarking" to address the need for validation skills.[7]

However, challenges remain in maintaining the balance between traditional and digital instruction.[42] While digital tools offer multimedia content and real-time assessments, they must not replace the critical thinking and "hands-on" learning experiences that are vital for a well-rounded engineering education.[42]

Mathematical Foundations: The Hidden Gaps

A final area of poorly explained topics resides in the advanced mathematical foundations required for CEM. Students frequently struggle with the "abstract nature" of electromagnetics, which is compounded by a disconnect from physical understanding.[10]

Specifically, topics like **Complex Analysis** and **Tensor Formalism** are identified as significant hurdles.[48] While complex analysis is essential for electronics, optics, and solving differential

equations, it is often treated as an elective rather than integrated into the core CEM curriculum.[48] Tensor formalism, critical for General Relativity and textured space models, is often taken for granted by professors, leaving students to "struggle heavily" with the transition from undergraduate to graduate-level research.[48]

Furthermore, the "Source Concept"—evaluating antenna and scattering characteristics solely through contrast currents—is a new paradigm that requires sophisticated modal decomposition and eigenvalue problem solving.[50] This method offers deep physical insight and reduces complexity, but it is rarely covered in introductory materials, which still rely heavily on direct inversion of system matrices.[50]

Conclusions and Strategic Recommendations

The identification of gaps in computational electromagnetics educational material reveals a discipline that is technically advanced but pedagogically fragmented. The "Black-Box" syndrome, combined with a lack of computational literacy and a disconnect from industrial needs, threatens to stifle innovation in high-stakes areas like 5G/6G, IoT, and quantum computing.

Key Conclusions for Educational Reform

The following conclusions synthesize the primary market research findings into a call for "Computational Thinking" in CEM:

- **Computational Literacy as a Core Competency:** Education must shift from software operation to algorithmic understanding, high-performance computing, and parallelization.[7]
- **Integration of Physics with AI:** AI-driven CEM must move beyond black-box surrogate modeling toward physics-informed networks that respect Maxwell's equations.[8]
- **Pedagogical Creativity and Visualization:** To attract the next generation of engineers, educational materials must leverage 3D animations, mobile apps, and visual laboratories to make the abstract tangible.[10]
- **Formal V&V and Benchmarking:** Verification and Validation skills must be embedded in the curriculum to ensure the reproducibility and reliability of simulations.[7]
- **Quantum-Ready Workforce:** Curricula must bridge the gap for classically-trained engineers to enter the field of Quantum Electromagnetics through potental-based formulations and quantum Hamilton equations of motion.[32]

By addressing these gaps, the electromagnetics community can foster a workforce that possesses both the theoretical depth of the "Founding Fathers" and the computational prowess required for the digital future. This will not only bridge the "Valley of Death" between academia and industry but also ensure that electromagnetics remains a vibrant and essential field of interest for the benefit of humanity.[6]

Works cited

1. (PDF) Computational Electromagnetics and the IEEE Antennas and Propagation Society: Seventy-five years of shared history, part 1 - ResearchGate, accessed February 22, 2026, https://www.researchgate.net/publication/383075610_Computational_Electromagnetics_and_the_IEEE_Antennas_and_Propagation_Society_Seventy-five_years_of_shared_history_part_1
2. Computational Electromagnetics and the IEEE ... - IEEE Xplore, accessed February 22, 2026, https://ieeexplore.ieee.org/iel8/74/10712740/10633853.pdf
3. Guest editorial: Recent advances in computational electromagnetics - R Discovery, accessed February 22, 2026, https://discovery.researcher.life/article/guest-editorial-recent-advances-in-computational-electromagnetics/3b652ecdcdc238adb0a19fa57554bf6a
4. Science of Computational Electromagnetics; a journey through its intellectual challenges, accessed February 22, 2026, https://ece.engin.umich.edu/event/science-of-computational-electromagnetics-a-journey-through-its-intellectual-challenges
5. Bridging the Gap Between Academia and Industry - ATD, accessed February 22, 2026, https://www.td.org/content/atd-blog/bridging-the-gap-between-academia-and-industry
6. Bridging the Gap Between Academia and Industry In The Energy Sector | Climate Change AI, accessed February 22, 2026, https://www.climatechange.ai/blog/2022-10-24-bridge-gap-energy
7. Antennas and Propagation Magazine IEEE - Volume 67 - Issue 2 - Scribd, accessed February 22, 2026, https://www.scribd.com/document/868523922/Antennas-and-Propagation-Magazine-IEEE-Volume-67-Issue-2
8. Machine Learning Advances in Computational Electromagnetics | Request PDF, accessed February 22, 2026, https://www.researchgate.net/publication/373538946_Machine_Learning_Advances_in_Computational_Electromagnetics
9. How can I self-learn Finite Element Analysis : r/AskEngineers - Reddit, accessed February 22, 2026, https://www.reddit.com/r/AskEngineers/comments/bhs9n2/how_can_i_selflearn_finite_element_analysis/
10. Teaching and Learning Electromagnetic Plane Wave Reflection and Transmission Using 3D TV [Education Corner] | Request PDF - ResearchGate, accessed February 22, 2026, https://www.researchgate.net/publication/332216508_Teaching_and_Learning_Electromagnetic_Plane_Wave_Reflection_and_Transmission_Using_3D_TV_Education_Corner
11. IEEE Antennas Propagation Magazine AP Magazine, accessed February 22, 2026, https://ieeeaps.org/publications/ieee-antennas-and-propagation-magazine

12. Difficulties in teaching electromagnetism: an eight year experience at Pierre and Marie Curie University - ResearchGate, accessed February 22, 2026, https://www.researchgate.net/publication/272717795_Difficulties_in_teaching_electromagnetism_an_eight_year_experience_at_Pierre_and_Marie_Curie_University
13. Computational Electromagnetics Online, accessed February 22, 2026, https://empossible.net/
14. A leap-frog discontinuous Galerkin time-domain method of analyzing electromagnetic scattering problems - Chinese Physics B, accessed February 22, 2026, https://cpb.iphy.ac.cn/article/2017/1907/cpb_26_10_104101.html
15. Simulation of optical resonators using DGTD and FDTD - ResearchGate, accessed February 22, 2026, https://www.researchgate.net/publication/228714615_Simulation_of_optical_resonators_using_DGTD_and_FDTD
16. Prismatic discontinuous Galerkin time domain method with an integrated generalized dispersion model for efficient optical metasurface analysis - Optica Publishing Group, accessed February 22, 2026, https://opg.optica.org/abstract.cfm?uri=ome-10-10-2542
17. The Finite Integration Technique for Electromagnetics - NAFEMS, accessed February 22, 2026, https://www.nafems.org/publications/resource_center/knowledge-base-cemwg/finite-integration/
18. Electromagnetic Simulation Software Comparison: Nullspace EM vs. HFSS vs. CST Studio Suite, accessed February 22, 2026, https://www.nullspaceinc.com/resources/electromagnetic-simulation-software-comparison-nullspace-em-vs-hfss-vs-cst-studio-suite
19. An Equivalent Modeling Method for Electromagnetic Radiation of PWM Fans with Multiple Radiation Sources - MDPI, accessed February 22, 2026, https://www.mdpi.com/2076-3417/15/6/2887
20. Some Selected Unsolved Problems in Classical and Quantum Electromagnetics - PIER Journals, accessed February 22, 2026, https://www.jpier.org/ac_api/download.php?id=24072910
21. Teaching – Computational Electromagnetics (CEM) – TU Darmstadt, accessed February 22, 2026, https://www.cem.tu-darmstadt.de/teach/courses/index.en.jsp
22. Lecture 36 Computational Electromagnetics, Numerical Methods - Purdue Engineering, accessed February 22, 2026, https://engineering.purdue.edu/wcchew/ece604f20/Lecture%20Notes/Lect36.pdf
23. Modeling Challenging EMC Problems Using the Method of Moments - White Rose Research Online, accessed February 22, 2026, https://eprints.whiterose.ac.uk/id/eprint/115293/1/IEEE_EMC_Magazine_Paper_CONCEPT_II_Workshop.pdf
24. ECE 408 | Electrical & Computer Engineering - Illinois, accessed February 22, 2026, https://ece.illinois.edu/academics/courses/ece408
25. Parallel Processing Applied to Computational Electromagnetics - DTIC, accessed February 22, 2026, https://apps.dtic.mil/sti/tr/pdf/ADA242372.pdf

26. An Introduction to the Fundamentals of Mesh Generation in Finite Element Analysis - FEA Academy, accessed February 22, 2026, https://www.fea-academy.com/pdf/FEA%20Academy%20-%20The%20Fundamentals%20of%20Mesh%20Generation%20in%20Finite%20Element%20Analysis.pdf
27. CEM Workshop Lectures (3/11): Mesh Generation in CEM | PPTX - Slideshare, accessed February 22, 2026, https://www.slideshare.net/slideshow/cem-workshop-lectures-311-mesh-generation-in-cem/239718753
28. Mesh generation for computational analysis - IET Digital Library, accessed February 22, 2026, https://digital-library.theiet.org/doi/pdf/10.1049/cae.1986.0050?download=true
29. Non-Matching Mesh Generation - Emergent Mind, accessed February 22, 2026, https://www.emergentmind.com/topics/non-matching-mesh-generation-strategy
30. A Posteriori Error Estimation and Adaptive Discretization Refinement Using Adjoint Methods in CEM: A Study with a One - Colorado State University, accessed February 22, 2026, https://www.engr.colostate.edu/~notaros/Papers/TAP_Adjoints_FEM_1D.pdf
31. Mesh generation | Applications of Scientific Computing Class Notes - Fiveable, accessed February 22, 2026, https://fiveable.me/applications-of-scientific-computing/unit-8/mesh-generation/study-guide/ufKZVhMjwE3IVrrh
32. Quantum Electromagnetics: A New Look—Part I (Invited Paper) - ResearchGate, accessed February 22, 2026, https://www.researchgate.net/publication/308806109_Quantum_Electromagnetics_A_New_Look-Part_I_Invited_Paper
33. Short Courses, Tutorials and Workshops - IEEE AP-S/URSI 2025, accessed February 22, 2026, https://2025.apsursi.org/short_courses.php
34. Survey Confirms Technology Pressures and Skills Gaps in ..., accessed February 22, 2026, https://set.kellyservices.us/resource-center/survey-confirms-technology-pressures-and-skills-gaps-in-engineering
35. Rigorous Indoor Wireless Communication System Simulations With Deep Learning-Based Radio Propagation Models | Request PDF - ResearchGate, accessed February 22, 2026, https://www.researchgate.net/publication/386595149_Rigorous_Indoor_Wireless_Communication_System_Simulations_with_Deep_Learning-based_Radio_Propagation_Models
36. Read "An Assessment of the Divisions of the Physical Measurement Laboratory at the National Institute of Standards and Technology Located in Boulder, Colorado: Fiscal Year 2023" at NAP.edu, accessed February 22, 2026, https://www.nationalacademies.org/read/27338/chapter/10
37. Quantum Electromagnetics: A New Look, Part II - Index of /, accessed February 22, 2026, https://jontalle.web.engr.illinois.edu/Public/ChewNewLookQED2.pdf

38. Call for Nominations for the 2025 IEEE Antennas and Propagation Society Field Awards, accessed February 22, 2026, https://ieeexplore.ieee.org/iel8/74/10807734/10807772.pdf
39. (PDF) Bridging the Skills Gap: Examining the Electronics Technology Students' Competence and Industry Demands - ResearchGate, accessed February 22, 2026, https://www.researchgate.net/publication/383889594_Bridging_the_Skills_Gap_Examining_the_Electronics_Technology_Students'_Competence_and_Industry_Demands
40. Tackling the electrical engineering skills gap through new ways of working, accessed February 22, 2026, https://blog.se.com/services/2025/04/18/addressing-the-electrical-engineering-skills-shortage/
41. K-12 Instruction Material Market Size & Forecast to 2030 - Research and Markets, accessed February 22, 2026, https://www.researchandmarkets.com/report/k-12-instruction-material-market
42. K-12 Instruction Material - MarketResearch.com, accessed February 22, 2026, https://www.marketresearch.com/Global-Industry-Analysts-v1039/Instruction-Material-42681402/
43. Education Market Research Reports | PreK-12, Higher Ed, & Digital Publishing Industry Trends - The Freedonia Group, accessed February 22, 2026, https://www.freedoniagroup.com/reports-analysis/education
44. Three Things We Love About EdReports' 'State of the Instructional Materials Market' Report, accessed February 22, 2026, https://www.cemd.org/three-things-we-love-about-edreports-state-of-the-instructional-materials-market-report/
45. IEEE Journal of Selected Topics in Electromagnetics, Antennas and Propagation (JSTEAP), accessed February 22, 2026, https://ieeeaps.org/ieee-jsteap
46. Best YouTube channel/ free learning for electromagnetics? : r/rfelectronics - Reddit, accessed February 22, 2026, https://www.reddit.com/r/rfelectronics/comments/1gbkhnz/best_youtube_channel_free_learning_for/
47. Best YT channel to learn Electromagnetics? : r/ElectricalEngineering - Reddit, accessed February 22, 2026, https://www.reddit.com/r/ElectricalEngineering/comments/1n2fq1q/best_yt_channel_to_learn_electromagnetics/
48. What was your hardest undergrad physics course - Reddit, accessed February 22, 2026, https://www.reddit.com/r/Physics/comments/1aykc79/what_was_your_hardest_undergrad_physics_course/
49. What advanced math branches do physicists use other than the usual diff geometry, topology and hypergeometry? - Reddit, accessed February 22, 2026, https://www.reddit.com/r/math/comments/1e0zud9/what_advanced_math_branches_do_physicists_use/
50. Computational Electromagnetics (CEM) - Katedra elektromagnetického pole

Fakulta elektrotechnická, accessed February 22, 2026, https://elmag.fel.cvut.cz/oblasti-vyzkumu/cem/
51. IEEE AP-S Field Interest Antennas Propagation, accessed February 22, 2026, https://ieeeaps.org/news/our-field-of-interest